

Class-Conditional Sharpness-Aware Minimization for Deep Long-Tailed Recognition

Zhipeng Zhou^{1,†}, Lanqing Li^{2,4,†}, Peilin Zhao^{3,*}, Pheng-Ann Heng², Wei Gong^{1,*}

¹University of Science and Technology of China, ²The Chinese University of Hong Kong,

³Tencent AI Lab, ⁴Zhejiang Lab

zzp1994@mail.ustc.edu.cn, lanqingli1993@gmail.com, masonzhao@tencent.com,

pheng@cse.cuhk.edu.hk, weigong@ustc.edu.cn

Abstract

It's widely acknowledged that deep learning models with flatter minima in its loss landscape tend to generalize better. However, such property is under-explored in deep long-tailed recognition (DLTR), a practical problem where the model is required to generalize equally well across all classes when trained on highly imbalanced label distribution. In this paper, through empirical observations, we argue that sharp minima are in fact prevalent in deep long-tailed models, whereas naïve integration of existing flattening operations into long-tailed learning algorithms brings little improvement. Instead, we propose an effective two-stage sharpness-aware optimization approach based on the decoupling paradigm in DLTR. In the first stage, both the feature extractor and classifier are trained under parameter perturbations at a class-conditioned scale, which is theoretically motivated by the characteristic radius of flat minima under the PAC-Bayesian framework. In the second stage, we generate adversarial features with class-balanced sampling to further robustify the classifier with the backbone frozen. Extensive experiments on multiple long-tailed visual recognition benchmarks show that, our proposed Class-Conditional Sharpness-Aware Minimization (CC-SAM), achieves competitive performance compared to the state-of-the-arts. Code is available at <https://github.com/zzpustc/CC-SAM>.

1. Introduction

Modern deep learning models, composed of multiple neural network layers with millions of parameters, have achieved remarkable successes in computer vision [24, 33,

41, 48]. A key enabler of deep learning is the collection of large-scale datasets [29, 42, 64], which are normally split into training and testing sets with presumably i.i.d. samples. However, such scenario provides relatively trivial tests for the generalization of machine learning models. In practice, label [17, 25, 56] and domain [14, 18, 23] distribution shifts are prevalent, due to the disparity between the data preparation and evaluation protocols. A classical example is imbalanced [15] or long-tailed recognition [60], where a model is trained on highly imbalanced source label distribution $p_s(\mathbf{y})$ while evaluated on a uniform target label distribution $p_t(\mathbf{y})$.

In this paper, we focus on the practical yet challenging deep long-tailed recognition (DLTR) problem, which is inherent in the visual world [32, 60] with fundamental connections to many disciplines such as the power-law scaling in network science [2] and the Pareto principle in economics [39]. In computer vision, numerous deep long-tailed learning studies have emerged in recent years, which mainly belong to 5 categories: class rebalancing [7, 10, 28, 40, 45, 52, 54], information augmentation [22, 27, 31, 50, 55], decoupled training [20, 58, 62], representation learning [9, 32, 51, 59, 65] and ensemble learning [6, 53, 63].

In this work, we propose a novel approach to DLTR from a *distinct angle*, by seeking out flat minima in the loss landscape of modern neural networks to ensure model robustness under parameter perturbation. Such optimization strategy, termed *flattening* in our context, have been shown in a myriad of literature to effectively improve generalization of deep learning models in terms of supervised learning [13, 21, 35, 43], self-supervised learning [30] and continual learning [11, 44]. However, application and adaption of flattening in the context of DLTR remain under-explored. To fill this gap, we first show later in this paper

[†] Equal contribution. ^{*} Corresponding authors. Work was primarily done when Z. Zhou worked as L. Li's intern at Tencent AI Lab, Shenzhen.

(Section 2.2.2) that existing flattening methods are ineffective for long-tailed learning, consistent with the observation from a very recent paper [49], due to severe label distribution shifts. Accordingly, we present a new efficient variant of the sharpness-aware minimization (SAM) [13] technique based on the Decoupling paradigm [20] of DLTR, which leverages the invariance of the class conditional distribution between the source and the target domain. In a nutshell, our contributions are three-fold:

- Through the lens of flattening, we corroborate a very recent observation [49] that existing sharpness-aware minimization techniques are suboptimal for deep long-tailed learning, justifying the need of more effective approaches to this important and practical issue.
- We introduce **Class-Conditional Sharpness-Aware Minimization (CC-SAM)**, a novel algorithm tailored for DLTR that improves model generalization by robust training against class-conditioned parameter perturbation. This technique is motivated by the characteristic radius of flat minima we derive under the PAC-Bayesian framework and can be implemented efficiently.
- We integrate CC-SAM with the two-stage decoupling paradigm of DLTR. Extensive experiments demonstrate that our method can achieve competitive performance on multiple public DLTR benchmarks, with remarkable robustness to out-of-distribution samples for open long-tailed recognition [32].

2. Preliminaries

2.1. Problem Setup

Throughout the paper, we denote scalars as s , vectors as $\mathbf{s}$, sets as $\mathcal{S}$ and equality by definition as $:=$. For a typical d -way classification task in DLTR, suppose we are given a training dataset $\mathcal{S} = \bigcup_{i=1}^n \{(\mathbf{x}_i, \mathbf{y}_i)\}$, where n is the total number of samples. Let's denote $\{n_1, n_2, \dots, n_d\}$ as the sample number of each class, where $n = \sum_i n_i$. Without loss of generality, we assume $n_i < n_j$ if $i < j$, and usually $n_d \gg n_1$, following a highly imbalanced class distribution. We further write the training (source) distribution as $p_s(\mathbf{x}, \mathbf{y}) = p_s(\mathbf{x}|\mathbf{y})p_s(\mathbf{y})$ and the testing (target) distribution as $p_t(\mathbf{x}, \mathbf{y}) = p_t(\mathbf{x}|\mathbf{y})p_t(\mathbf{y})$. Consider a family of models parameterized by $\mathbf{w} \in \mathcal{W} \subseteq \mathbb{R}^k$; given a loss function $l: \mathcal{W} \times \mathcal{X} \times \mathcal{Y} \rightarrow \mathbb{R}_+$, we define the empirical training loss $L_S(\mathbf{w}) := \frac{1}{n} \sum_{i=1}^n l(\mathbf{w}, \mathbf{x}_i, \mathbf{y}_i)$ and the population testing loss $L_T(\mathbf{w}) := \mathbb{E}_{(\mathbf{x}, \mathbf{y}) \sim p_t(\mathbf{x}, \mathbf{y})} [l(\mathbf{w}, \mathbf{x}, \mathbf{y})]$. Having observed only $\mathcal{S}$, the goal is to optimize for model parameters $\mathbf{w}$ having lowest risk $L_T(\mathbf{w})$ at test time.

In DLTR, a key challenge is the label distribution shift between the training and testing sets, i.e., $p_s(\mathbf{y}) \neq$

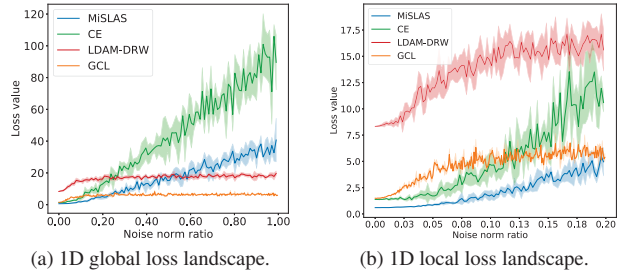


Figure 1. Loss value vs. noise norm ratio $\|\epsilon\|_2/\|\theta\|_2$. All experiments are conducted on the CIFAR-10-LT with an imbalance ratio of 100, implemented with the same backbone ResNet-32 and averaged over 5 random seeds.

$p_t(\mathbf{y})$ [17]. However, it's logical to assume *no distribution shift* within each individual class. That is, the training and testing samples are drawn i.i.d. from the same class-conditional distribution, which indicates $p_s(\mathbf{x}|\mathbf{y}) = p_t(\mathbf{x}|\mathbf{y})$. Here, we build our flattening algorithm upon the decoupling paradigm of DLTR [20], where the feature extractor $f(\mathbf{x}; \varphi)$ and classifier $h(\mathbf{z}; \theta)$ are trained by distinct sampling strategies in a two-stage manner, with $\mathbf{w} := (\varphi, \theta)$.

2.2. Motivation

Previous studies have shown strong connection between the geometry of the loss landscape and generalization of deep learning models [1, 16, 19, 21]. In this section, we first empirically demonstrate that many deep long-tailed models exhibit sharp minima, and then show that naïve application of classical flattening operations results in limited improvement. Motivated by these observations, we theoretically analyze the characteristic radius of flat minima in DLTR to show that a class-conditional flattening procedure is required.

2.2.1 Sharpness in Deep Long-Tailed Models

To show the pervasive sharpness in current DLTR models, we select two baselines (empirical risk minimization of cross-entropy (CE), LDAM-DRW [7]) and two advanced methods (MiSLAS [62], GCL [26]) from recent literature to conduct experiments¹. By perturbing the parameter θ of the classifier h with increasing noise ϵ , we observe the loss of each model on CIFAR-10-LT in Figure 1.

Figure 1(a) shows that, at global scale, the loss of CE is the sharpest across a wide noise range. However, due to the fact that DLTR methods usually employ distinct loss functions, it's hard to compare the sharpness of their loss landscapes in a fair and meaningful way. Moreover, large noise ratio can easily flip the sign of parameters and deteriorate

¹We re-implement all models via their publicly released code.

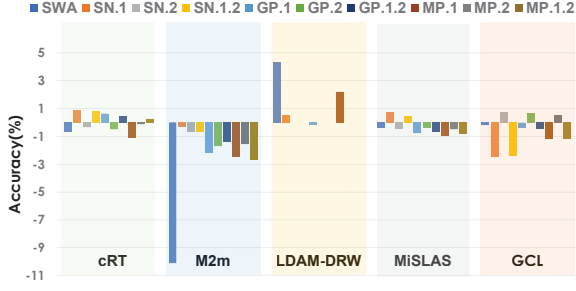


Figure 2. Performance gain of naïve integration of flattening operations with DLTR algorithms. The number represents the stage where flattening applies. For example, “SN.1.2” means we apply spectral normalization on both stages. Only the first-stage result of LDAM is reported since it’s a one-stage approach. All experiments are conducted on CIFAR-100-LT with imbalance ratio of 100.

the classifier decision boundary. Therefore in this paper, we only consider locally flat minima, i.e., in the perturbative ($\|\epsilon\|_2 \ll \|\mathbf{w}\|_2$) regime shown in Figure 1(b). From this local view, *only MiSLAS* is observed to have a comparable or flatter minimum than CE, suggesting that the existing DLTR methods have poor generalization and robustness against model parameter perturbation. More detailed 2D loss landscapes covering overall, head and tail classes are depicted in **Appendix E**, and corroborate our observation in 1D.

2.2.2 Naïve Integration of Flattening Procedures

To improve the generalization of deep long-tailed models, we experiment to navigate flatter minima by integrating the existing flattening operations (stochastic weight averaging (SWA) [16], spectral normalization (SN) [4], gradients penalization (GP) [61], and model perturbation (MP) [44]) into five representative deep long-tailed learning baselines to see if they are generally beneficial for DLTR. Illustrated in Figure 2, these flattening methods bring little or negative gains in most cases. Even though in some cases they are beneficial, the corresponding baselines (mainly LDAM-DRW) are too weak so that the final performance are still sub-optimal, which calls for more effective flattening procedures for DLTR. More implementation details are provided in the **Appendix C**.

2.2.3 Characteristic Radius of Flat Minima

Let’s re-consider flattening operations in DLTR from a theoretical perspective. The empirical loss $L_S(\mathbf{w})$ is typically non-convex for deep neural network models, whose landscape may exhibit multiple local or global minima with similar values of $L_S(\mathbf{w})$ while having drastically different generalization performance (i.e., significantly different values

of $L_T(\mathbf{w})$). In order to find a more effective approach for generalization in DLTR, motivated by the high correlation between sharpness of the loss landscape and model generalization [19], we follow the sharpness-aware minimization (SAM) framework [13] by optimizing an upper bound of $L_T(\mathbf{w})$ derived from the PAC-Bayesian framework [34]. Given a prior over the parameters before observing any data and a posterior over the parameters dependent on the training set and learning algorithm, the PAC-Bayesian framework bounds the generalization error of any model in terms of the KL divergence between the two probability distributions. Assuming a Gaussian prior and posterior, we arrive at the following generalization bound (formal proof in **Appendix D**):

Theorem 1 (*Perturbative PAC-Bayesian Generalization Bound*). For any $\rho > 0$, $0 < \delta < 1$, number of samples $n \in \mathbb{N}^+$, $k := \dim(\mathbf{w})$, with probability at least $1 - \delta$ over a training set $\mathcal{S}$ sampled i.i.d. from distribution $\mathcal{T}$,

$$L_T(\mathbf{w}) \leq \max_{\|\epsilon\|_2 \leq \sqrt{k}\rho} L_S(\mathbf{w} + \epsilon) + \sqrt{\frac{\frac{\|\mathbf{w}\|_2^2}{4\rho^2} + \log(\frac{n}{\delta}) + \mathcal{O}(1)}{n-1}}.$$

Intuitively, optimizing the perturbative bound as a surrogate function effectively seeks out parameter solutions whose neighborhoods have uniformly low empirical training loss, thus achieving a flat minimum. Now, our key insight is that the bound is convex with respect to the radius ρ of the flat minima, hence there exists an optimal ρ^* which gives the tightest generalization bound:

$$\rho^* := \arg \min_{\rho} \left[\max_{\|\epsilon\|_2 \leq \sqrt{k}\rho} L_S(\mathbf{w} + \epsilon) + \sqrt{\frac{\frac{\|\mathbf{w}\|_2^2}{4\rho^2} + \log(\frac{n}{\delta})}{n-1}} \right]. \quad (1)$$

We denote ρ^* as the *characteristic radius* of the flat minima centered at $\mathbf{w}$. It’s worth mentioning that such optimal perturbation radius is overlooked by previous literature on sharpness-aware optimization, mainly due to arguments that the PAC-Bayesian bound is too loose to capture the real generalization error [35]. However, our Theorem 1 inspired from [8, 12] ensures a non-vacuous bound (i.e., the square root term falls below 1, see empirical evidence in **Appendix B**) in the over-parameterized “deep learning” regime, rendering the characteristic radius relevant for optimization.

Moreover, the perturbative bound can be approximated by a first-order Taylor expansion, yielding

$$\begin{aligned} \max_{\|\epsilon\|_2 \leq \sqrt{k}\rho} L_S(\mathbf{w} + \epsilon) &\approx \max_{\|\epsilon\|_2 \leq \sqrt{k}\rho} [L_S(\mathbf{w}) + \epsilon^T \nabla_{\mathbf{w}} L_S(\mathbf{w})] \\ &= L_S(\mathbf{w}) + \sqrt{k}\rho \|\nabla_{\mathbf{w}} L_S(\mathbf{w})\|_2 \end{aligned} \quad (2)$$

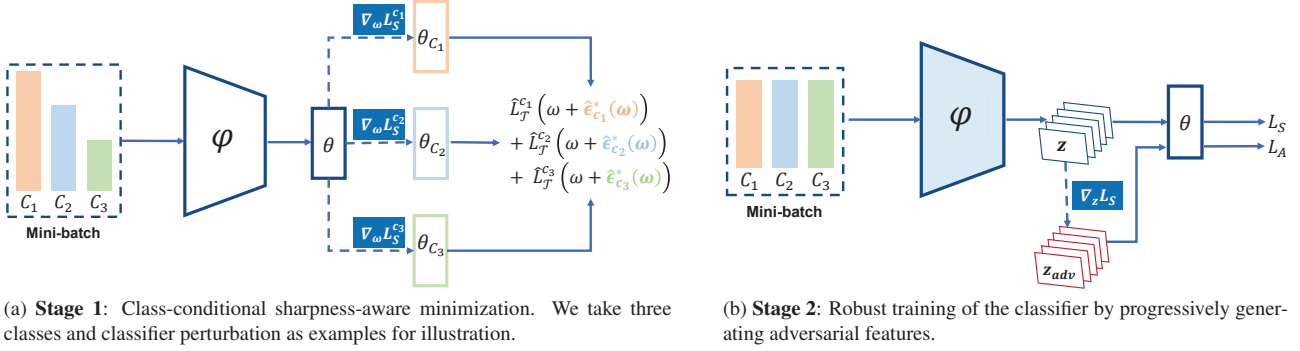


Figure 3. The overall framework of CC-SAM.

and the optimal perturbation vector

$$\begin{aligned}\hat{\epsilon}^*(\mathbf{w}) &\approx \arg \max_{\|\epsilon\|_2 \leq \sqrt{k}\rho} [L_S(\mathbf{w}) + \epsilon^T \nabla_{\mathbf{w}} L_S(\mathbf{w})] \\ &= \sqrt{k}\rho \frac{\nabla_{\mathbf{w}} L_S(\mathbf{w})}{\|\nabla_{\mathbf{w}} L_S(\mathbf{w})\|_2}.\end{aligned}\quad (3)$$

Substitute Eqn 2 into 1, and ignore $\log(\frac{n}{\delta})$ by assuming $\rho \ll \|\mathbf{w}\|_2$ as we discussed in Sec 2.2.1, obtaining

$$\rho^* \approx \left(\frac{\|\mathbf{w}\|_2}{2\|\nabla_{\mathbf{w}} L_S(\mathbf{w})\|_2} \right)^{\frac{1}{2}} k^{-\frac{1}{4}} (n-1)^{-\frac{1}{4}} \quad (4)$$

with the approximated optimal generalization bound

$$\hat{L}_{\mathcal{T}}(\mathbf{w}) \approx \max_{\|\epsilon\|_2 \leq \sqrt{k}\rho^*} L_S(\mathbf{w} + \epsilon) + \frac{1}{2\sqrt{n-1}} \cdot \frac{\|\mathbf{w}\|_2}{\rho^*}. \quad (5)$$

The first term captures the sharpness of L_S at $\mathbf{w}$, and the second term serves as a regularizer on the magnitude of $\mathbf{w}$, akin to the L2 regularization in SAM [13]. A key difference here is that our regularization term is well motivated by theoretic interpretation from the PAC-Bayesian bound.

3. Methodology

In this section, we introduce our two-stage sharpness-aware optimization algorithm based on Decoupling [20] in detail. In the first stage of the decoupled training, both the feature extractor and classifier are trained under parameter perturbations at a class-conditioned scale. In the second stage, we generate adversarial features to further robustify the classifier while freezing the backbone. The complete pseudo-code is shown in Algorithm 1.

3.1. Stage 1: Class-Conditional Sharpness-Aware Minimization (CC-SAM)

Note that a key assumption we made in Theorem 1 when deriving the characteristic radius of flat minima is that the

training data $\mathcal{S}$ are drawn i.i.d. from the target distribution $\mathcal{T}$. However, as we explained in Sec 2.1, such assumption is invalid due to the severe label distribution shift between training and testing sets in DLTR. To resolve this fundamental conflict, we turn to the class conditional distribution $p_s(\mathbf{x}|\mathbf{y})$ and $p_t(\mathbf{x}|\mathbf{y})$ instead, for which the i.i.d. assumption reasonably holds. Accordingly, we decompose the total loss $L_{\mathcal{T}}(\mathbf{w})$ and derive a generalization bound for each class separately, obtaining

$$L_{\mathcal{T}}(\mathbf{w}) = \sum_{c=1}^k L_{\mathcal{T}}^c(\mathbf{w}) = \sum_{c=1}^k \mathbb{E}_{\mathbf{x} \sim p_t(\mathbf{x}|c)} [l(\mathbf{w}, \mathbf{x}, c)], \quad (6)$$

$$\begin{aligned}\hat{L}_{\mathcal{T}}^c(\mathbf{w}) &= \max_{\|\epsilon\|_2 \leq \sqrt{k}\rho_c^*} L_S^c(\mathbf{w} + \epsilon) + \frac{1}{2\sqrt{n-1}} \cdot \frac{\|\mathbf{w}\|_2}{\rho_c^*}, \\ &\approx (2\|\mathbf{w}\|_2 \|\nabla_{\mathbf{w}} L_S^c(\mathbf{w})\|_2)^{\frac{1}{2}} k^{\frac{1}{4}} (n_c-1)^{-\frac{1}{4}}\end{aligned}\quad (7)$$

$$\rho_c^* = \left(\frac{\|\mathbf{w}\|_2}{2\|\nabla_{\mathbf{w}} L_S^c(\mathbf{w})\|_2} \right)^{\frac{1}{2}} k^{-\frac{1}{4}} (n_c-1)^{-\frac{1}{4}}, \quad (8)$$

$$\hat{\epsilon}_c^*(\mathbf{w}) \approx \sqrt{k}\rho_c^* \frac{\nabla_{\mathbf{w}} L_S^c(\mathbf{w})}{\|\nabla_{\mathbf{w}} L_S^c(\mathbf{w})\|_2}. \quad (9)$$

An immediate observation of Eqn 8 is that the characteristic radius ρ_c^* is class-dependent, more specifically, negatively correlated with the label frequency n_c . One possible interpretation is that the model is more confident regarding the majority classes with higher n_c , due to non-asymptotic estimation error given limited samples, hence it requires a smaller perturbative region in the parameter space to ensure flat loss landscape with respect to that class. Additionally, the class-wise generalization bound $\hat{L}_{\mathcal{T}}^c(\mathbf{w})$ in Eqn 7 is positively related to the class-wise gradient norm $\|\nabla_{\mathbf{w}} L_S^c(\mathbf{w})\|_2$, which effectively enforces hard example mining.

It follows that optimizing the approximated generalization bound $\hat{L}_{\mathcal{T}}(\mathbf{w}) = \sum_{c=1}^k \hat{L}_{\mathcal{T}}^c(\mathbf{w})$ yields an effective algorithm in the first stage, namely Class-Conditional

Algorithm 1: Training Paradigm of CC-SAM

Input: Training Dataset

$$\mathcal{S} \sim p_s(\mathbf{x}, \mathbf{y}) = p_s(\mathbf{x}|\mathbf{y})p_s(\mathbf{y})$$

Output: Model trained with CC-SAM**Stage 1:**Initialize $\mathbf{w} = \{\varphi, \theta\}$ randomly**while** not converged **do** **foreach** batch $\mathcal{B}_i$ in $\mathcal{S}$ **do** Compute empirical loss $L_{\mathcal{S}}$ with $\mathcal{B}_i$

Estimate the class-specific gradient set

 $G = \{\mathbf{g}_1, \mathbf{g}_2, \dots, \mathbf{g}_k\}$ with respect to $L_{\mathcal{S}}$ Perturb $\mathbf{w}$ with G according to Eqn 8 and Eqn 9 Update $\mathbf{w}$ via Eqn 10 and Eqn 11**Stage 2:**Freeze φ **while** not converged **do**

Sample batches via class-balanced sampler

$$\mathcal{S}' = \{\mathcal{B}'_1, \mathcal{B}'_2, \dots, \mathcal{B}'_m\} \sim p_s(\mathbf{x}|\mathbf{y}) \cdot \text{Uniform}(\mathbf{y})$$

foreach batch $\mathcal{B}'_i$ in $\mathcal{S}'$ **do** Computing empirical loss $L_{\mathcal{S}}$ with $\mathcal{B}'_i$

Obtain the adversarial feature via Eqn 12

Evaluate overall loss according to Eqn 13 and Eqn 14

 Update θ accordingly

Sharpness-Aware Minimization (CC-SAM):

Given learning rate η ,

$$\mathbf{w} \leftarrow \mathbf{w} - \eta \nabla_{\mathbf{w}} L_{\mathcal{S}}^{\text{CC-SAM}}(\mathbf{w}) \quad (10)$$

$$\approx \mathbf{w} - \eta \sum_{c=1}^k \nabla_{\mathbf{w}} \hat{L}_{\mathcal{T}}^c(\mathbf{w})|_{\mathbf{w}+\hat{\mathbf{e}}_c^*(\mathbf{w})}, \quad (11)$$

which is computationally efficient since it only involves first-order gradients. Even when $\mathbf{w}$ is close to the optimal $\mathbf{w}^*$, where $\nabla_{\mathbf{w}} L_{\mathcal{S}}(\mathbf{w})|_{\mathbf{w}^*} \approx 0$, the gradient for each class $\nabla_{\mathbf{w}} L_{\mathcal{S}}^c(\mathbf{w})|_{\mathbf{w}^*}$ for estimating the optimal perturbation vector $\hat{\mathbf{e}}_c^*(\mathbf{w})$ in Eqn 9 is most likely far from zero, circumventing the need for computing high-order terms.

3.2. Stage 2: Robust Training of the Classifier

For the second stage, we freeze the backbone to maintain feature representation and concentrate on refining the decision boundary of the classifier. Following Decoupling, we adopt the class-balanced sampling strategy in this stage, to rectify the classifier with uniform label distribution. As depicted in the Figure 3(b), when taking original feature $\mathbf{z}$ as the input of classifier $h(\mathbf{z}; \theta)$, we can generate adversarial

features in forward pass as follows:

$$\mathbf{z}_{adv} = \mathbf{z} + \lambda \frac{\nabla_{\mathbf{z}} L_{\mathcal{S}}(\mathbf{z}; \theta)}{\|\nabla_{\mathbf{z}} L_{\mathcal{S}}(\mathbf{z}; \theta)\|_2} \quad (12)$$

$$L_{\mathcal{A}} = h(\mathbf{z}_{adv}; \theta) \quad (13)$$

where λ is the hyper-parameter to scale the adversarial gradient $\nabla_{\mathbf{z}} L_{\mathcal{S}}(\mathbf{z}; \theta) / \|\nabla_{\mathbf{z}} L_{\mathcal{S}}(\mathbf{z}; \theta)\|_2$. Moreover, we employ a progressive strategy to balance $L_{\mathcal{S}}$ and $L_{\mathcal{A}}$. At epoch t ,

$$L = (1 - \frac{t}{T})L_{\mathcal{S}} + \frac{t}{T}L_{\mathcal{A}} \quad (14)$$

where T is the total number of epochs in this stage, and $L_{\mathcal{A}}$ dominates the training progressively.

4. Evaluation

In this section, we evaluate our method on multiple mainstream DLTR datasets and compare it with popular baselines including the state-of-the-art (SOTA) methods. Moreover, we also report the result on open-set recognition [32, 47] tasks to demonstrate the excellent robustness of CC-SAM to out-of-distribution samples. In the end, an ablation study is presented to verify the effectiveness of our design choices.

4.1. Datasets and Baselines

Following the mainstream evaluation protocol [7, 63], we conduct experiments on five major long-tailed datasets, CIFAR-10-LT, CIFAR-100-LT, Places-LT [32], ImageNet-LT [32], and iNaturalist 2018 [46].

CIFAR-10-LT/CIFAR-100-LT: These two datasets are sampled from the original CIFAR with different imbalance ratios $\beta = N_{max}/N_{min}$, where N_{max} and N_{min} are the corresponding number of the most and least frequent classes. Following [26], we set the imbalance ratio as {200, 100, 50} for evaluation.

Places-LT & ImageNet-LT: Both datasets were first proposed by OLTR [32]. Places-LT contains 62.5K training images spanning 365 classes in total, with imbalance ratio 996. ImageNet-LT has 115.8K training images covering 1000 categories, with imbalance ratio being 256.

iNaturalist 2018: As a naturally long-tailed classification dataset, iNaturalist 2018 contains 437.5K training images from 8142 categories, and its imbalance ratio is 512. We follow the official split in our evaluations.

For fair comparison, we exclude ensemble or pre-training models in our experiments. The baselines and state-of-the-art methods evaluated include (1) class rebalancing: LDAM-DRW [7], LDAM-DRW + SAM [38], De-confound-TDE [45], Lifted Loss [36], Focal Loss [28], OpenMax [3], BBN [63], LADE [17], DisAlign [58],

	CIFAR-10-LT			CIFAR-100-LT		
Imbalance Ratio	200	100	50	200	100	50
CE	65.68	70.70	74.81	34.84	38.43	43.90
CE + Mixup [57]	65.84	72.96	79.48	35.84	40.01	45.16
LDAM-DRW [7]	73.52	77.03	81.03	38.91	42.04	47.62
De-confound-TDE [45]	-	80.60	83.60	-	44.15	50.31
CE + Mixup + cRT [20]	73.06	79.15	84.21	41.73	45.12	50.86
BBN [63]	73.47	79.82	81.18	37.21	42.56	47.02
Contrastive Learning [51]	-	81.40	85.36	-	46.72	51.87
BGP [49]	-	-	-	41.20	45.20	50.50
MiSLAS [62]	77.31	82.06	85.16	42.33	47.50	52.62
VS + SAM [38]	-	82.40	-	-	46.60	-
GCL [26]	<u>79.03</u>	<u>82.68</u>	<u>85.46</u>	<u>44.88</u>	<u>48.71</u>	<u>53.55</u>
CC-SAM	80.94	83.92	86.22	45.66	50.83	53.91

Table 1. Top-1 Accuracy on CIFAR-10-LT and CIFAR-100-LT. All methods take ResNet-32 as the backbone. All baseline results except BGP are directly adopted from [26]. “-” means the original paper didn’t report the corresponding results.

cRT	Stage 1 + dir	Stage 1 + mag	Stage 2	Acc
✓				37.8
✓	✓			38.9
✓		✓		37.5
✓	✓	✓		40.1
✓	✓	✓	✓	40.6

Table 2. Ablation studies on Places-LT. “Stage 1 + dir” enforces parameter perturbation along the recommended direction ($\nabla_w L_S^c(\mathbf{w}) / \|\nabla_w L_S^c(\mathbf{w})\|_2$ in Eqn 9) with magnitude of 1, whereas “Stage 1 + mag” enforces perturbation with the recommended magnitude (ρ_c^* in Eqn 9) in a random direction.

LUNA [5], BGP [49], VS+SAM [38] (2) information augmentation: RSG [50], (3) decoupled training: Decouple- τ -norm [20], cRT [20], MiSLAS [62], GCL [26], (4) representation learning: Range Loss [59], OLTR [32], IEM [65], Contrastive Learning [51], ResLT [9].

Our code is implemented with Pytorch 1.4.0 and all experiments are carried out on Tesla V100 GPUs. We train each model with batch size of 64 (for CIFAR-10-LT and CIFAR-100-LT) / 128 (for Places-LT) / 256 (for ImageNet-LT) / 512 (for iNaturalist 2018), SGD optimizer with momentum of 0.9. We apply similar tricks in Balanced Softmax [40] as the mainstream methods have done. Although we implemented CC-SAM under a two-stage framework, it serves as a general technique to improve the existing DLTR methods without model perturbation (Appendix A.3).

Intuitively, CC-SAM brings more computation overhead due to additional gradient descents. But we only perturbed the last several layers as an efficient version of CC-SAM in

our evaluation. A simple training time comparison is presented in Appendix A.5.

4.2. Main Results

We evaluate CC-SAM on the five mainstream public benchmarks mentioned above, and the corresponding results are reported in Table 1 and Table 3. The best method is bolded and the second best is underlined. According to the evaluations, we observe that CC-SAM unanimously attains the best ranking on CIFAR-LT datasets. On large scale datasets (Places-LT, ImageNet-LT, and iNaturalist 2018), CC-SAM shows competitive performance compared to other advanced methods, too. Specifically, it consistently brings better gains to the medium and tail classes, demonstrating the effectiveness of class-conditional flattening for deep long-tailed recognition.

4.3. Open Long-Tailed Recognition

The ability to detect out-of-distribution samples from the open world, namely open-set recognition [47], provides a unique dimension for evaluating the robustness of deep learning models. Open-set long-tailed recognition, or OLTR [32], combines the open-set problems with deep long-tailed learning to offer even more challenging tasks for model generalization. Following the setting of OLTR [32], we enable CC-SAM to distinguish the close-set and open-set samples by applying a simple, non-learnable prototype-based metric. Results are presented in Table 4. We find that CC-SAM achieves the top F-measure performance and outperforms LUNA [5] on Places-LT, a SOTA with a sophisticated open-set detection method based on hierarchical metrics. The experiments demonstrate the remarkable gen-

Dataset	Method	Backbone	Many	Medium	Few	Overall
ImageNet-LT	CE	ResNeXt-50	65.9	37.5	7.7	44.4
	Decouple- τ -norm [20]	ResNet-50	56.6	44.2	27.4	46.7
	Balanced Softmax [40]	ResNeXt-50	64.1	48.2	33.4	52.3
	LADE [17]	ResNeXt-50	64.4	47.7	34.3	52.3
	RSG [50]	ResNeXt-50	63.2	48.2	32.2	51.8
	DisAlign [58]	ResNet-50	61.3	52.2	31.4	52.9
		ResNeXt-50	62.7	52.1	31.4	53.4
	ResLT [9]	ResNeXt-50	63.0	<u>53.3</u>	35.5	52.9
	BGP [49]	ResNet-50	-	-	-	51.5
	MiSLAS [62]	ResNet-50	-	-	-	52.7
	LDAM-DRW + SAM [38]	ResNet-50	62.0	52.1	34.8	53.1
	GCL [26]	ResNet-50	-	-	-	<u>54.9</u>
	CC-SAM	ResNet-50	61.4	49.5	<u>37.1</u>	52.4
		ResNeXt-50	63.1	53.4	41.1	55.4
Places-LT	CE	ResNet-152	45.7	27.3	8.2	30.2
	Decouple- τ -norm [20]	ResNet-152	37.8	40.7	31.8	37.9
	Balanced Softmax [40]	ResNet-152	42.0	39.3	30.5	38.6
	LADE [17]	ResNet-152	42.8	39.0	31.2	38.8
	RSG [50]	ResNet-152	41.9	41.4	<u>32.0</u>	39.3
	DisAlign [58]	ResNet-152	40.4	42.4	30.1	39.3
	ResLT [9]	ResNet-152	39.8	43.6	31.4	39.8
	MiSLAS [62]	ResNet-152	-	-	-	40.2
	GCL [26]	ResNet-152	-	-	-	<u>40.6</u>
	CC-SAM	ResNet-152	41.2	42.1	36.4	40.6
iNaturalist 2018	CE	ResNet-50	72.2	63.0	57.2	61.7
	Decouple- τ -norm [20]	ResNet-50	65.6	65.3	65.9	65.6
	Balanced Softmax [40]	ResNet-50	-	-	-	70.6
	LADE [17]	ResNet-50	-	-	-	70.0
	RSG [50]	ResNet-50	-	-	-	70.3
	DisAlign [58]	ResNet-50	-	-	-	70.6
	ResLT [9]	ResNet-50	-	-	-	70.2
	BGP [49]	ResNet-50	<u>70.0</u>	69.9	69.6	70.5
	MiSLAS [62]	ResNet-50	-	-	-	<u>71.6</u>
	LDAM-DRW + SAM [38]	ResNet-50	64.1	<u>70.5</u>	<u>71.2</u>	70.1
	GCL [26]	ResNet-50	-	-	-	72.0
	CC-SAM	ResNet-50	65.4	70.9	72.2	70.9

Table 3. Top-1 Accuracy on Places-LT, ImageNet-LT and iNaturalist 2018. As for Places-LT, we take a pre-trained ResNet-152 as the backbone for a fair comparison.

eralization of CC-SAM due to its capacity to learn highly robust representations.

4.4. Ablation Study

Since CC-SAM consists of multiple components, here we provide an ablation study to demonstrate their effectiveness. The result of experiments conducted on Places-LT are shown in Table 2. In particular, CC-SAM degrades to cRT without any of our proposed operations, which we take as a baseline. It is observed that CC-SAM benefits from the

design choices of each individual stage, and the integration of both stages shows the best performance.

We also study the class-conditional perturbation by differentiating the impacts of choosing the right direction $\nabla_w L_S^c(w)/\|\nabla_w L_S^c(w)\|_2$ and the right magnitude ρ_c^* in Eqn 9, shown as "stage 1 + dir" and "stage 1 + mag" in Table 2 respectively. For the former, we choose a perturbation scale of 1. For the latter, we perturb the model parameters by a random direction sampled from a zero-mean Gaussian. It turns out the perturbation direction contributes more, and

	ImageNet-LT				Places-LT			
Method	Many	Medium	Few	F-measure	Many	Medium	Few	F-measure
CE	40.1	10.4	0.4	0.295	45.9	22.4	0.4	0.366
Lifted Loss [36]	34.8	29.3	17.4	0.374	41.0	35.2	23.8	0.459
Focal Loss [28]	35.7	29.3	15.6	0.371	41.0	34.8	22.3	0.453
Range Loss [59]	34.7	29.4	17.2	0.373	41.0	35.3	23.1	0.457
OpenMax [3]	35.8	30.0	17.6	0.368	41.1	35.4	23.2	0.458
OLTR [32]	41.9	33.9	17.4	0.474	44.6	36.8	25.2	0.464
IEM [65]	46.1	42.3	20.1	0.525	48.8	42.4	28.9	0.486
LUNA [5]	48.2	44.7	23.6	0.579	48.1	41.6	29.0	<u>0.491</u>
CC-SAM	61.4	49.5	37.1	<u>0.552</u>	41.2	41.8	36.4	0.510

Table 4. Open long-tail performance of top-1 accuracy on ImageNet-LT and Places-LT. F-measure is a balanced treatment of precision and recall. The backbone of CC-SAM is ResNet-50 for ImageNet-LT, while other compared methods are equipped with ResNet-10.

applying both attains the best improvement. For more ablation studies, please refer to **Appendix A.1**.

5. Related Work and Discussion

5.1. Deep Long-Tailed Recognition

A latest survey [60] systematically studies up-to-date DLTR algorithms published at the top-tier conferences and introduces a new taxonomy, dividing DLTR methodologies into three categories: class re-balancing, information augmentation, and module improvement. For class re-balancing, a branch of sampling strategies [20, 40, 52, 54] have been proposed to re-balance the training distribution. Meanwhile, some studies design dedicated loss functions [7, 10, 28, 45] to dynamically re-weight the gradients for optimization. As one representative of information augmentation, RSG [50] enriches the feature space of tail classes via dynamical prototypes and a maximized vector loss. Decoupled training [20, 58, 62] propose to learn effective representation and robust classifier according to different sampling strategies in two stages.

Our method follows the decoupled training paradigm. Moreover, to combat label distribution shift, our method seeks out flat minima by perturbing the model parameters at a class-conditional scale, which is unseen in existing DLTR literature.

5.2. Flattening for Deep Learning Models

Sharpness as a generalization measure of deep learning models has been extensively explored in previous literature [1, 13, 16, 19]. Keskar *et al.* [21] empirically found that a large batch SGD optimizer might lead to sharp minima and poor generalization. Inspired by this, He *et al.* [16] implicitly averaged along the SGD trajectory to find the asymmetric valleys, namely asymmetric flat minima. Due to the power of flattening, it has also been applied to many

learning tasks to achieve better generalization. Rangwani *et al.* [37] introduced a domain adversarial training framework by adding an auxiliary loss to smooth the local minima for domain adaptation. Shi *et al.* [44] demonstrated that a flattened model was resilient to catastrophic forgetting in class-incremental few-shot learning.

In this work, we investigate flattening in the context of DLTR. To our best knowledge, there are only a few works which succeed in applying flattening to improve generalization in DLTR/OLTR. GBP [49] is built upon the observation of high correlation between logits and gradient norm, and proposes a gradient penalty loss as regularization. However, it lacks rigorous theoretical interpretation by employing a very loose (vacuous) PAC-Bayesian bound and meanwhile under-performs compared to our approach (Table 1, 3). Liu *et al.* [30] develops a variant of SAM that re-weights the perturbation at instance level to combat data imbalance, without specifying an optimal perturbation or giving any theoretical justification; thus, we regard it as a simplistic and incomplete version of our method. Besides, Rangwani *et al.* [38] observes that re-weighting techniques in DLTR lead to saddle points for tail classes, and directly leverages SAM to alleviate this problem. In contrast, CC-SAM further advances [38] by providing a theoretically motivated class-conditional SAM design for better generalization and robustness to label distribution shift.

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